Buying tips for Soundcards

■ Define your usage. If you are an avid music lover, go for a soundcard that supports multiple channels (like 4.1 surround). If you want a gaming solution, buy a card that supports the standard 3D sound APIs, like DirectSound, EAX, A3D and I3DL2.

■ The vendor should provide a minimum of a one-year warranty for budget soundcards (under Rs 2,000) and at least a three-year warranty for more expensive ones.

■ Before buying, make sure that the soundcard's interface is PCI or USB and not ISA. If you plan to use your soundcard for playing games or watching DVD movies, get a PCI soundcard.

■ Check that the soundcard you are buying has drivers for your favourite operating system. Windows users should not face problems, but the software required for Linux, for example, might be hard to come by.

■ If you are an audiophile and care about things like signal loss, buy a soundcard that has gold-plated connectors on it.

■ Check to see if the soundcard comes bundled with a decent variety of applications (such as games and media players). If you are a first time buyer, these will prove very useful to get you started with your soundcard.

Buying tips for Speakers

■ **Power rating:** In most speaker systems available today, the power rating, in watts, is deceptively specified as Peak Music Power Output (PMPO). This particular specification is nothing but marketing hype—you should always go by the RMS power of a speaker system as this is the true technical measure of the speaker's capability to handle continuous power. For surround sound speaker systems, a rating of 40 Watts RMS is a good place to start for your computer-based surround speaker system.

■ **Frequency response:** If your speaker system has a frequency range that is limited at the upper end (for example, if it cannot perform above 16 KHz), you will not hear any of the high treble, and will lose definition in your music, especially if you listen to a lot of classical music. Similarly, if your speakers can't go below 50 Hz, they won't be able to render deep bass well, like drums, and explosions in movies.

■ **Stands:** If you have 2.1 speakers that are primarily meant for music playback, you won't need to have speaker stands as you would probably place them right on your desk. However, if you have a 4.1 or 5.1 speaker system, you would need to ensure that they either have stands on all the satellites, or a base attached to these speakers. This is useful if you want to place them behind you on a wall, or even standing on the floor at ear-level for proper surround sound imaging.

■ **Controls:** Speakers with an in-line volume control give you the freedom to control the master speaker volume without having to reach troublesome locations like the subwoofer where the volume controls are generally placed.

■ **Audio inputs:** If you have a 2.1 stereo speaker system, you only need standard analogue left and right channels. However, in the case of a four-point surround sound system, you would need to look out for four separate channels of analogue audio inputs to your speaker system. This would be in the form of two 3.5 mm stereo inputs (left/right for front and left/right for rear), or four separate RCA inputs. Also watch out for pseudo-four channel speakers that are basically just stereo, with two speakers for each side. In the case of 5.1 speakers, you would either need to have six separate RCA inputs for the discrete AC-3 channels (if the signal is already being split by an external decoder like in the Sound Blaster Live! 5.1 DE soundcard), or you would need a DIN connector that is fed by the digital output of your soundcard in order to have true Dolby Digital rendering in your surround sound system.

Soundcard Connecters Demystified